GENETIC ALGORITHM FOR LOW-COST WATER FILTRATION SYSTEMS

A Data-Driven Optimization Approach Supporting SDG 6 Clean Water and Sanitation

PROJECT REPORT

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Course : Machine Learning

Course Code: ITA0604

ABSTRACT

Access to safe and affordable drinking water remains a major development challenge in many rural and underserved communities. Filtration systems can reduce selected physical, chemical, and microbiological contaminants, but their effectiveness depends on the media used, layer thickness, flow rate, maintenance schedule, and local water characteristics. A design that is inexpensive may not be safe, while a high-performance design may be unaffordable or difficult to maintain. This project proposes a data-driven approach that uses a Genetic Algorithm (GA) to search for practical low-cost filter configurations.

A Genetic Algorithm is a population-based optimization method inspired by natural selection. Each candidate filter design is encoded as a chromosome containing design variables such as sand depth, gravel depth, activated-carbon quantity, filter area, and operating flow rate. Candidate designs are evaluated using a fitness or objective function that combines lifecycle cost, water-quality penalties, flow-rate penalties, and maintenance considerations. The optimization is subject to hard constraints: designs that fail required drinking-water safety criteria are rejected rather than being allowed to compensate for unsafe water through lower cost.

The proposed workflow includes problem definition, data collection, cleaning and preparation with Pandas, numerical analysis with NumPy, GA-based optimization, visualization, and laboratory validation. Charts can show cost versus performance, fitness convergence across generations, and comparisons among feasible designs. The report also considers local material availability, user needs, durability, environmental impact, and the importance of transparent documentation.

The project is a design and analysis framework rather than a claim that a particular filter has already been proven safe. Illustrative values must be replaced with locally collected prices and measured water-quality data. Any selected configuration must be tested by a qualified laboratory against applicable drinking-water standards before use. The work supports United Nations Sustainable Development Goal 6, especially Target 6.1, by exploring how optimization can contribute to safe, affordable, and reliable drinking-water access.

Keywords: Genetic Algorithm, water filtration, optimization, NumPy, Pandas, fitness function, SDG 6.

1. INTRODUCTION

1.1 Background

Clean drinking water is essential for health, education, livelihoods, and community development. Water sources may contain suspended particles, microorganisms, dissolved chemicals, or naturally occurring contaminants. The type and level of contamination vary by region and season, so no single filter design is suitable for every source. In resource-constrained settings, a filtration system must be affordable, easy to operate, repairable with locally available materials, and capable of producing an adequate quantity of water.

1.2 Need for Optimization

Traditional design often relies on expert experience and repeated physical trials. These methods are useful but may take time and may not explore many combinations of materials and dimensions. Optimization can help compare numerous possible configurations systematically. A Genetic Algorithm is suitable when the design contains mixed variables, nonlinear relationships, and competing objectives such as cost, performance, flow, and maintenance.

1.3 Project Aim

The aim is to develop a conceptual, data-driven optimization framework that identifies candidate filter designs with low lifecycle cost while meeting defined water-quality and operational constraints.

1.4 Objectives

- Identify relevant design variables and performance measures for a low-cost filter.
- Prepare a structured dataset of materials, prices, laboratory results, flow rates, and maintenance needs.
- Use NumPy and Pandas to clean, analyze, and summarize the data.
- Define a fitness function and apply Genetic Algorithm operators to search the design space.
- Visualize the trade-offs among cost, water quality, and flow rate.
- Map the project to SDG 6 and specify validation, safety, and documentation requirements.

1.5 Scope

The project focuses on optimizing the configuration of a small-scale filtration unit. It does not assume that filtration alone removes every dissolved chemical or pathogen. Disinfection, source protection, safe storage, and specialist treatment may be necessary depending on the water source and measured contaminants.

2. PROBLEM UNDERSTANDING

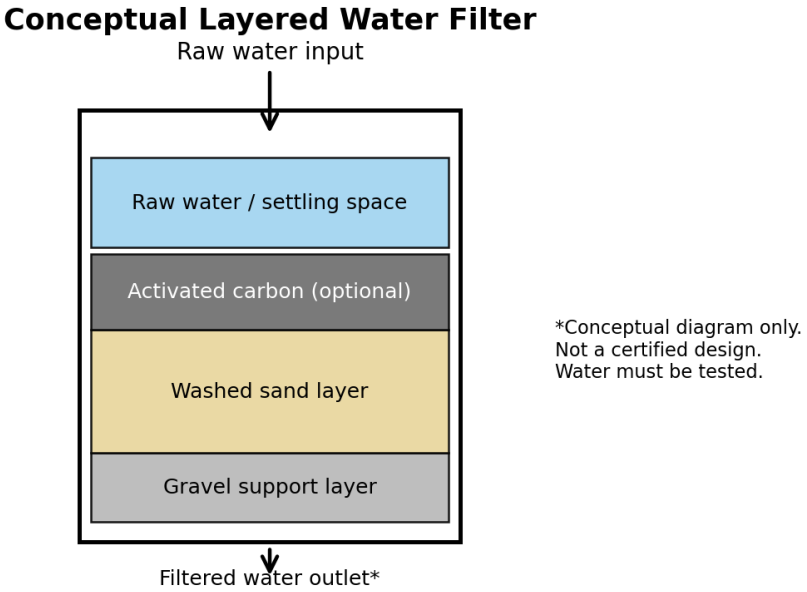


Figure 1. Conceptual filter layers; the actual media configuration requires testing.

2.1 Problem Statement

Communities with limited resources need water-treatment options that are affordable, reliable, and appropriate for local conditions. The design challenge is to select filter media and dimensions that reduce relevant contaminants and deliver sufficient water without making the system too expensive to build, operate, or maintain.

2.2 Stakeholders

- Households and community users who need safe water and simple operation.
- Local technicians who install, inspect, and maintain the system.
- Public-health and water-quality officers who define safety requirements.
- Local suppliers who provide filter media and replacement parts.
- Researchers and engineers who test configurations and analyze performance.

2.3 Design Variables

A candidate solution may encode sand-layer depth, gravel-layer depth, activated-carbon mass, filter area, flow-control setting, and media replacement interval. Variables should be bounded by practical construction limits. For example, a layer cannot have negative thickness, and the total filter depth must fit within the selected container.

Variable	Example unit	Reason to include
Sand depth	cm	Influences particle capture and hydraulic resistance
Gravel depth	cm	Supports the media and affects drainage
Activated carbon	kg	May reduce selected organic

		compounds and improve taste/odour
Flow rate	L/hour	Determines the amount of water produced
Replacement interval	days/months	Affects lifecycle cost and performance consistency

2.4 Constraints and Assumptions

The design must meet locally applicable water-quality limits, minimum output requirements, structural stability, and maintenance feasibility. Material performance must be based on measurements or credible technical evidence. Assumptions are recorded explicitly so the model does not treat estimated values as verified facts.

3. DATA COLLECTION

3.1 Data Required

A reliable optimization process depends on representative data. The dataset should connect each tested configuration with its material quantities, purchase cost, water-quality measurements before and after treatment, flow rate, pressure loss where relevant, and maintenance observations.

- Material data: name, grade, local availability, unit price, and expected service life.
- Water-quality data: raw and treated measurements for contaminants relevant to the source.
- Operational data: flow rate, operating duration, clogging observations, and cleaning frequency.
- Lifecycle data: installation cost, replacement costs, labour, transport, and disposal.
- Context data: seasonal source variation, household demand, climate, and user preferences.

3.2 Potential Sources

Data may be collected through controlled laboratory experiments, local supplier quotations, municipal or public water-quality reports, technical publications, and interviews with water-system operators. Sources should be dated and documented. Supplier prices should include currency, location, and purchase quantity because costs vary across regions.

3.3 Experimental Plan

1. Select representative raw-water samples and measure baseline quality.
2. Prepare candidate filters with clearly recorded media type and dimensions.
3. Run tests under consistent operating conditions and record flow rate.
4. Measure relevant contaminants in influent and effluent using appropriate methods.
5. Repeat tests to estimate variability and record failures or maintenance needs.
6. Store raw observations separately from cleaned analysis data.

3.4 Data Quality

Measurements should include units, sampling date, instrument or laboratory method, detection limits where relevant, and quality-control notes. Missing values should not automatically be replaced with averages when doing so could conceal unsafe results. The dataset should distinguish measured, estimated, and simulated values.

4. DATA PREPARATION

4.1 Cleaning and Validation

Data preparation converts collected records into a consistent table. Duplicate records are checked, unit formats are standardized, invalid values are flagged, and missing measurements are investigated. Values outside physical or instrument limits should be reviewed rather than silently deleted.

4.2 Feature Selection

Features should be selected based on the project goal and available evidence. Useful inputs include media quantities, layer dimensions, filter area, raw-water characteristics, and flow setting. Outcomes include treated-water quality, flow rate, maintenance frequency, and total lifecycle cost. Variables that do not affect the decision may be removed, but safety-critical measurements must not be omitted merely to simplify the model.

4.3 Train, Validation, and Test Separation

If a predictive model is trained to estimate filtration performance, data should be separated into training, validation, and test sets. Measurements from the same physical test run should not be split across sets, as this can cause data leakage. The final test set should be used only for the final performance check.

4.4 Example Dataset Structure

Design ID	Sand (cm)	Carbon (kg)	Cost (₹)	Flow (L/h)
D001	20	0.20	450	3.0
D002	25	0.30	560	2.6
D003	18	0.15	390	3.5

The entries above are fictional examples to illustrate data structure only. They are not validated designs and do not demonstrate that the water is safe. Each row should also include contaminant measurements, test conditions, maintenance data, and a record of whether safety criteria were met.

5. NUMPY AND PANDAS ANALYSIS

5.1 Role of Pandas

Pandas provides DataFrame structures for loading, inspecting, cleaning, grouping, and summarizing observations. It can calculate average costs by material combination, identify missing measurements, and compare performance across design IDs.

5.2 Role of NumPy

NumPy supports efficient numerical calculations, array operations, random-number generation, and normalization. It can be used to calculate cost totals, normalized penalties, and candidate fitness values within a GA.

5.3 Illustrative Python Workflow

```
import pandas as pd
import numpy as np

# Example only: replace with verified experimental data.
df = pd.read_csv("filter_trials.csv")

# Inspect the dataset and identify missing measurements.
print(df.head())
print(df.isna().sum())

# Keep a record of data quality checks before analysis.
required = ["cost_rupees", "flow_lph", "quality_penalty"]
analysis = df.dropna(subset=required).copy()

# Example summary statistics.
print(analysis[required].describe())

# Example normalized cost and penalty (guard against zero ranges).
cost_range = analysis["cost_rupees"].max() - analysis["cost_rupees"].min()
if cost_range > 0:
    analysis["cost_norm"] = (
        (analysis["cost_rupees"] - analysis["cost_rupees"].min()) / cost_range
    )
else:
    analysis["cost_norm"] = 0.0

print(analysis[["cost_rupees", "cost_norm"]].head())
```

5.4 Interpretation

The code demonstrates a simple analysis pipeline; it is not a complete water-safety model or a complete Genetic Algorithm. The quality penalty must be defined from the contaminants relevant to the source and must reflect applicable standards. Data cleaning decisions should be justified and documented so that potentially unsafe observations are not hidden.

6. GENETIC ALGORITHM METHODOLOGY

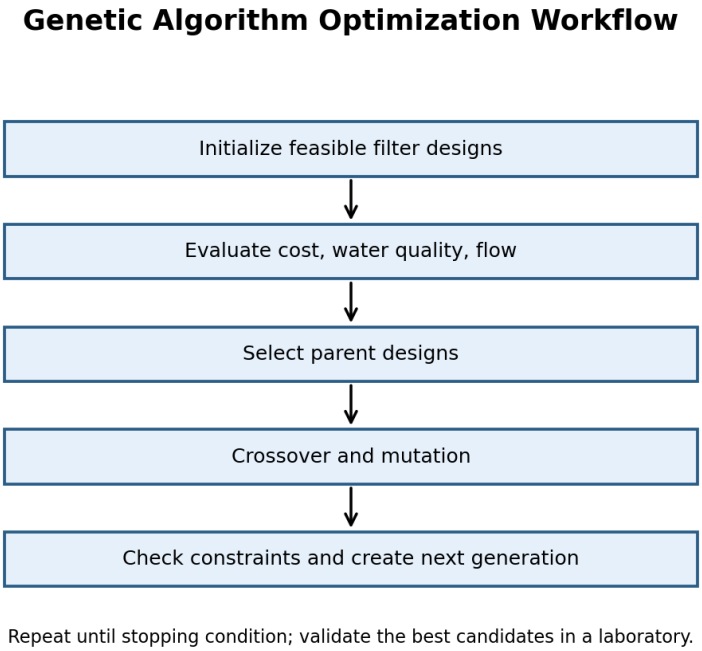


Figure 2. Main stages of the Genetic Algorithm search process.

6.1 Chromosome Representation

Each chromosome represents one candidate filter configuration. A simple chromosome might be [sand depth, gravel depth, carbon mass, filter area, target flow]. Discrete variables can represent available media types, while continuous variables represent quantities or dimensions. The representation must prevent physically impossible combinations.

6.2 Initial Population

The algorithm begins with a population of randomly generated but feasible candidate designs. A mixture of random and expert-informed candidates can improve coverage while ensuring that early solutions respect basic construction limits.

6.3 Selection

Selection gives better-performing candidates a greater chance of becoming parents. Tournament selection is a common approach: a small random group is compared and the best feasible candidate is selected. Safety constraints are applied before ranking.

6.4 Crossover and Mutation

Crossover combines parts of two parent chromosomes to create new candidates. Mutation makes small random changes, such as increasing sand depth or changing carbon quantity. Mutation helps explore alternatives that may not appear through crossover alone. Repair rules or feasibility checks keep the new designs within valid limits.

6.5 Replacement and Stopping

A new generation is formed from offspring and, optionally, a few elite candidates. The algorithm stops after a specified number of generations, when improvement becomes small for several generations, or when an acceptable feasible solution is found. Several independent runs should be performed because GA results can vary with random initialization.

6.6 Pseudocode

```
Create an initial population of feasible filter designs
Evaluate safety constraints and objective score for each design
Repeat until stopping condition is met:
    Select parent designs from feasible candidates
    Apply crossover to create offspring
    Apply mutation to explore new configurations
    Repair or reject infeasible offspring
    Evaluate cost, water quality, flow, and maintenance
    Keep the best feasible candidates for the next generation
Return candidate designs for laboratory validation
```

7. FITNESS FUNCTION AND CONSTRAINTS

7.1 Why a Fitness Function Is Needed

A fitness function provides a consistent way to compare candidate designs. Since the project has multiple objectives, the function should account for cost, water-quality performance, flow rate, and maintenance. However, a weighted score must never allow low cost to compensate for failure of a critical drinking-water safety requirement.

7.2 Example Objective Function

For feasible designs, one possible minimization objective is:

$$\mathbf{J(x) = wc \times Cn(x) + we \times Qp(x) + wf \times Fp(x) + wm \times Mn(x)}$$

Here, Cn is normalized lifecycle cost, Qp is a normalized water-quality penalty, Fp is a penalty for output below the required flow, and Mn is a normalized maintenance burden. The weights wc, we, wf, and wm reflect the relative priorities for feasible designs.

7.3 Safety Constraints

- Relevant treated-water measurements must meet applicable drinking-water standards.
- Required output must meet a defined minimum under realistic operating conditions.
- Material choices must be appropriate for contact with drinking water.
- Structural, leakage, and maintenance requirements must be satisfied.
- Designs with missing critical safety results should be treated as unverified, not safe.

7.4 Feasibility-First Strategy

A robust strategy first rejects designs that fail hard safety or engineering constraints. The objective function is then used to rank the remaining feasible designs. This prevents an unsafe but cheap filter from receiving a good score. Where contaminant data are uncertain, conservative limits and additional testing should be used.

7.5 Lifecycle Cost

Total cost should include the container, media, labour, transport, water wasted during operation, cleaning, replacement parts, and disposal. A design with a low purchase price but frequent replacement may have a higher lifecycle cost than a more durable option.

8. VISUALIZATION AND INTERPRETATION

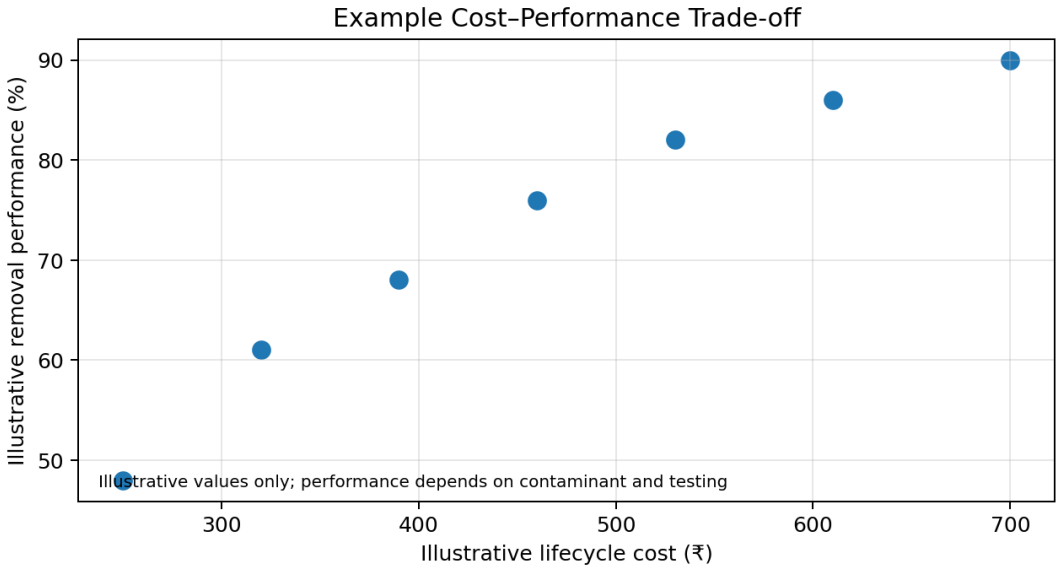


Figure 4. Illustrative cost–performance trade-off; values are not experimental results.

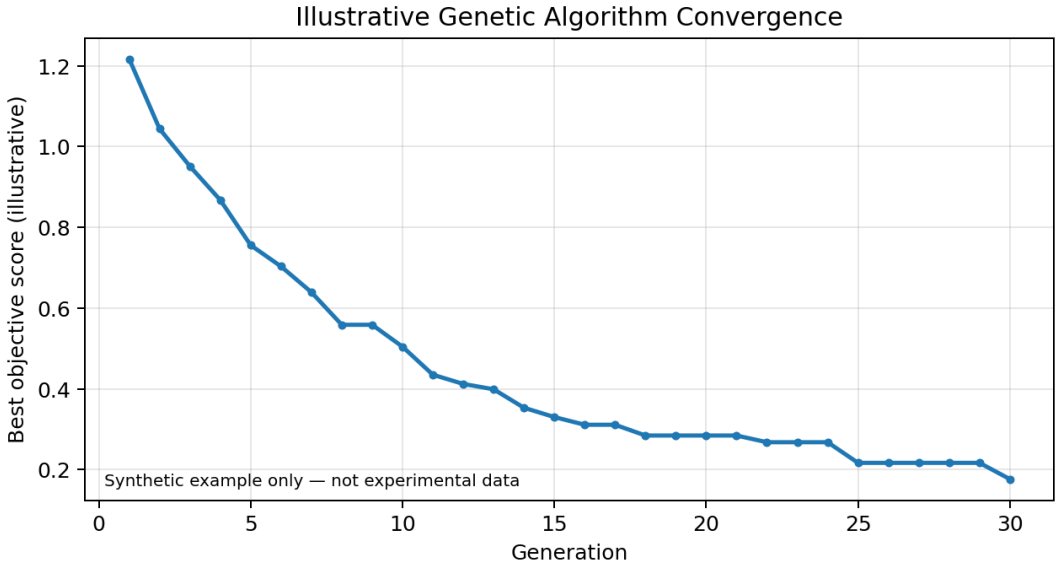


Figure 3. Illustrative convergence curve using synthetic values.

8.1 Fitness Convergence

A convergence graph plots the best feasible objective score against generation number. A downward trend indicates that the search is finding better candidates. A flat curve may indicate convergence, a limited search space, or insufficient diversity; it does not prove that the global optimum has been found.

8.2 Cost Versus Performance

A scatter plot can place lifecycle cost on the horizontal axis and a water-quality performance measure on the vertical axis. Feasible designs can be highlighted separately from unsafe or unverified designs. The chart helps decision-makers see whether small cost increases provide meaningful performance benefits.

8.3 Flow Rate Comparison

A bar chart can compare flow rate among candidate designs. High flow is not automatically better: excessive flow may reduce contact time or compromise removal performance. The flow target should be chosen from the treatment process and household demand.

8.4 Example Plotting Code

```
import matplotlib.pyplot as plt

# Assumes columns have been measured or calculated correctly.
plt.figure()
plt.plot(history["generation"], history["best_objective"])
plt.xlabel("Generation")
plt.ylabel("Best feasible objective score")
plt.title("Genetic Algorithm Convergence")
plt.tight_layout()
plt.show()

plt.figure()
plt.scatter(results["cost_rupees"], results["removal_percent"])
plt.xlabel("Lifecycle cost (₹)")
plt.ylabel("Measured removal (%)")
plt.title("Cost versus Measured Performance")
plt.tight_layout()
plt.show()
```

8.5 Interpretation Guidelines

Graphs must display units, sample sizes, and whether values are measured or simulated. Results should include uncertainty where possible. If removal percentages are used, the report must name the contaminant and test method because performance against one contaminant does not establish overall drinking-water safety.

9. INNOVATION AND SDG MAPPING

9.1 Innovation

The main innovation is the use of evolutionary search to compare many possible filter configurations systematically. The approach can incorporate local material prices, measured water-quality data, maintenance needs, and practical constraints in one decision framework.

- Local sourcing: prioritize materials that are available and serviceable in the community.
- Lifecycle thinking: evaluate total ownership cost rather than only initial purchase price.
- Adaptive design: rerun optimization when source-water conditions or prices change.
- Transparent selection: retain records of candidate designs, constraints, scores, and test results.
- Community-centered evaluation: consider ease of cleaning, replacement availability, and user training.

9.2 SDG 6 Connection

United Nations Sustainable Development Goal 6 aims to ensure availability and sustainable management of water and sanitation for all. Target 6.1 focuses on universal and equitable access to safe and affordable drinking water. An optimized low-cost filter may contribute to this goal if the design is validated, affordable in practice, and integrated with safe handling and maintenance.

9.3 Responsible Claims

The project can demonstrate alignment with SDG 6, but it should not claim that a prototype solves regional water access by itself. Access also depends on source protection, infrastructure, governance, affordability, user acceptance, and continuous quality monitoring.

9.4 Wider Benefits

A locally maintainable system can reduce dependence on expensive imported components and improve resilience. Yet the environmental benefits depend on media sourcing, replacement frequency, waste management, and the energy and water used during operation.

10. VALIDATION, SAFETY, AND ETHICS

10.1 Laboratory Validation

Optimization outputs are candidate designs, not proof of safe water. The selected configuration should be tested by a qualified laboratory using sampling and analytical methods appropriate to the source. Tests should cover the contaminants of concern and should be repeated under different operating conditions, including after media ageing or maintenance.

10.2 Treatment Limitations

Sand and gravel primarily support physical filtration and flow distribution. Activated carbon can reduce some organic compounds and improve taste or odour, but it does not reliably remove all dissolved chemicals or pathogens. Depending on source-water conditions, additional treatment such as validated disinfection or specialized contaminant removal may be required.

10.3 User Safety

- Do not distribute untreated or unverified water as safe for drinking.
- Provide clear instructions for installation, cleaning, media replacement, and safe storage.
- Test treated water at suitable intervals and after major changes or repairs.
- Use materials suitable for drinking-water contact and avoid contaminated media.
- Consult local public-health authorities when contamination is suspected.

10.4 Ethical and Social Considerations

The design should be accessible to users with limited income and should not shift hidden maintenance costs onto households. Data should be reported honestly, including failed trials and uncertainty. Community members should be consulted about usability, cultural needs, willingness to maintain the system, and the affordability of replacement media.

10.5 Risk Register

Risk	Potential impact	Mitigation
Incorrect or incomplete data	Misleading design choice	Repeat tests and document methods
Unsafe treated water	Health harm	Hard safety constraints and laboratory verification
Media unavailable locally	Downtime and higher cost	Check supply chain and alternatives
Filter clogging	Reduced output	Measure pressure loss and maintenance interval
Cost estimates outdated	Poor affordability	Refresh local quotations before deployment

11. IMPLEMENTATION PLAN AND EXPECTED OUTCOMES

11.1 Project Workflow

7. Define the source-water problem and relevant safety limits.
8. Identify design variables, operational requirements, and cost boundaries.
9. Collect material, laboratory, and maintenance data.
10. Clean and analyze the dataset using Pandas and NumPy.
11. Implement a Genetic Algorithm with feasibility checks.
12. Run multiple trials and record convergence and candidate designs.
13. Visualize trade-offs and select feasible candidates for testing.
14. Validate prototypes and revise the model using measured results.
15. Prepare documentation and recommendations for responsible use.

11.2 Suggested Timeline

Phase	Activity	Suggested duration
1	Problem definition and literature review	1 week
2	Data collection and experimental planning	2 weeks
3	Data preparation and exploratory analysis	1 week
4	GA implementation and debugging	2 weeks
5	Visualization and candidate comparison	1 week
6	Prototype testing and report completion	2 weeks

11.3 Expected Outcomes

- A structured dataset for filter design and performance.
- A reproducible NumPy/Pandas analysis workflow.
- A Genetic Algorithm that proposes candidate configurations.
- Plots showing fitness convergence and cost-performance trade-offs.
- A documented list of feasible candidates for laboratory validation.
- An SDG 6 mapping and a risk-aware implementation plan.

11.4 Evaluation Metrics

Evaluation should consider lifecycle cost, measured contaminant concentrations, removal efficiency for each target contaminant, litres produced per hour, maintenance frequency, durability, and user acceptability. The report should compare the GA-selected design with a baseline design under the same test conditions.

12. LIMITATIONS AND FUTURE ENHANCEMENTS

12.1 Limitations

The quality of the optimization depends on the quality and coverage of the data. If only a few filter configurations have been tested, a model may not accurately predict performance for untested combinations. Water sources vary seasonally, and a design that works for one source may fail for another. A weighted fitness function also requires careful choice of weights, while a Genetic Algorithm may return different results across runs.

12.2 Technical Limitations

- The GA may find a good candidate without proving it is globally optimal.
- Laboratory testing can be costly and time-consuming.
- Performance may change as media clog, age, or become contaminated.
- Some contaminants require specialized treatment beyond basic filtration.
- Cost and material availability may vary by location and over time.

12.3 Future Enhancements

- Use multi-objective optimization to identify a Pareto front for cost, quality, and flow.
- Include uncertainty intervals and robust optimization for seasonal source variation.
- Use surrogate models to estimate performance and reduce the number of physical trials.
- Add lifecycle environmental impact and waste disposal to the objective.
- Build a simple dashboard for entering local prices and comparing feasible designs.
- Conduct community field trials with local authorities and qualified laboratories.

12.4 Multi-Objective Alternative

Instead of combining all objectives into one weighted score, a multi-objective GA can identify designs for which no objective can be improved without worsening another. These non-dominated solutions form a Pareto front. Decision-makers can then choose among safe designs based on local affordability, water demand, and maintenance capacity.

13. DOCUMENTATION, CONCLUSION, AND REFERENCES

13.1 Documentation Checklist

- Problem statement, objectives, scope, and assumptions.
- Dataset source, units, collection dates, and quality-control records.
- Data-cleaning decisions and feature-selection rationale.
- Chromosome representation, GA settings, constraints, and stopping rule.
- Fitness-function definitions and justification of weights.
- Code version, random seeds, software versions, and reproducibility notes.
- Tables, plots, uncertainty discussion, failed trials, and laboratory results.
- Safety instructions, limitations, SDG mapping, and references.

13.2 Conclusion

This project presents a systematic method for exploring low-cost water filtration designs using Genetic Algorithms and data analysis with NumPy and Pandas. The GA represents filter configurations as chromosomes and improves them through selection, crossover, and mutation. A suitable objective function considers lifecycle cost, water-quality penalties, insufficient flow, and maintenance, while hard safety constraints prevent unsafe candidates from being selected merely because they are inexpensive.

Visualization makes the optimization process easier to interpret by showing convergence and trade-offs among cost, water quality, and output. The proposed workflow encourages local sourcing, maintainability, transparent data handling, and laboratory verification. Its contribution to SDG 6 is potential rather than automatic: a filter must be proven effective for the specific water source and supported by appropriate disinfection, safe storage, monitoring, and user training where needed.

The most important conclusion is that computational optimization can help engineers search efficiently, but it cannot replace water-quality testing or public-health safeguards. Future work should collect real local data, compare GA results with baseline designs, test durability, and assess affordability over the full service life.

13.3 References

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Note: Before final submission, verify reference details and add any local drinking-water standards, experimental papers, supplier quotations, and laboratory methods actually used in the project.

IMAGE NOTES

The figures in this report were created for educational illustration. The convergence and cost–performance plots use synthetic example values, not measured test results. The filter schematic is conceptual and does not certify that the arrangement makes water safe to drink. Actual filter designs must be selected based on source-water testing and validated by a qualified laboratory.